



dia. The cellular phone boom in India is estimated to be exploding at over 70 per cent, annually. "India's mobile telephone market is one of the fastest growing markets with increased penetration of mobile telephony in rural areas, resulting in further growth in telecom sector. Upgrade of the DL and RC from paper licence to smart card is going to witness huge demand. Many states have already adopted it and many others are to follow soon," states Deepak Agarwal, senior manager, technical marketing, Secure Micros, STMicroelectronics, India.

Experts estimate the Indian smart card industry growing at 45 per cent, annually. In the next five years, the Indian smart card population is expected to increase eight fold. Increasing use of this plastic card across various segments will fuel the volume of contactless smart card shipment with a CAGR of around 24 per cent during 2011-2013. At the applications front, financial services will witness rapid growth among other sectors like telecom, transport and public services. Rising online payment market through cards will be another main driver for growth in this sector.

With the *rashtriya swasthya bima yojana* (RSBY), National Rural Employment Guarantee Act (NREGA) and the public distribution system (PDS),

there is sure to be an upsurge in the adoption of smart card technology in India. "Under the programme for below poverty line (BPL) families, the government has already issued close to 180 million smart cards and there is a potential for over 300 million cards still to be issued/reissued to these families," says Anil Agarwal.

Banking is another sector where the demand for smart cards is rising as smart cards are slowly changing over from magnetic strip cards to being chip-based, or a combination of the two. "These chips, which are based on smart card technology, are Europay Mastercard Visa (EMV) certified and offer a high level of security for any banking transaction," says Deepak Agarwal. This vertical is also expected to witness a growth in smart card adoption in the years to come.

Similarly, ID verification is another field that will experience a growth in the demand for smart cards. With the aim to secure and safeguard identities against theft, this vertical will receive special attention. Also, the Delhi Metro Railway Corporation (DMRC) has been using the technology extensively. "There has been an enormous increase in the use of DMRC smart cards over the last two years. The demand rose from 0.45 million to 1.32 million from the first year of implementing the

technology, growing to 2.1 million. There has been almost a five fold increase in demand within a span of just three years," says P. Viswanatha Bhas, managing director, Siepmann's Card Systems. Siepmann offers all types of contactless smart cards, with various memory capacities.

Technology: An enabler

Technology is now becoming an enabler for bringing new services to the consumers and helping them to meet challenges. Originally, a smart card was nothing more than a memory chip, and had a simple mechanism. It was also easy for hackers to break the mechanism. Today's smart cards are more complex with more functionality and security mechanisms, and therefore, these are now equipped with microprocessors and a significant number of security measures.

Today, we are witnessing a wave of mobile transformation, with 3G and 4G/LTE under active development. This has also led to a series of major developments in the smart card industry, for example, with the introduction of large capacity smart cards, new form factors, near field communication (NFC), integrated smart cards and biometrics. "The key factor which will contribute to the future growth of contactless smart card market will be the rollout of NFC, which will be supported by the widespread adoption of contactless transportation and payment systems," states Abishek Lakhota.

Originally, processor cards used a monolithic software model—operating system and application functions were closely interwoven. Today, most smart cards use a more modular software design and also application separation, which enable flexible application design. Applications can be developed and loaded even after card manufacturing, or even post-issuance.

A smart card essentially comprises a secure microprocessor core and non-volatile memory. Of late, these components have undergone changes,